SCI-PTC

Correlated-Mode Cleaning and Detector Coefficients

Scientific Rationale

Scientific contract v0.1 – document revision r0.3

TolTEC Scientific Contract Program
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Status. Owner-approved scope; bounded Stage B r0.3 freeze-candidate revision. The scientific architecture and the base-v0.1 frozen-subspace projection decision are approved. Scientific authority is not yet frozen. Implementation conformity, representation/response fidelity, observational performance, and production readiness remain unassessed.

Document role. This is the standalone science-team rationale. The companion *Engineering Conformance Specification* contains the complete shared normative notation, definitions, equations, assumptions, requirements, and falsifiable predictions. This rationale explains that authority without reproducing its full formal register.

Contents

1 What PTC estimates – and what it cannot identify

1.1 The calibrated detector-time problem

PTC begins after raw-timestream conditioning and calibration. Its primary input is an admitted SCI-CAL detector-time matrix in the top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention, accompanied by complete RTC/CAL identity, validity, influence, response, and uncertainty state. It is not raw readout coordinate, a map, or an arbitrary matrix with a familiar filename.

The organizing scientific model is

$$Y^{\mathrm{CAL}} = S + U_\star + N, \qquad U_\star = M_\star A_\star^{\mathsf{T}},$$

where S is astronomical signal, $U_\star$ is the latent correlated-mode or supplied-template component admitted by the signal model, and N is the remaining detector-noise term. PTC estimates $\widehat{U} = \widehat{M}\widehat{A}^{\mathsf{T}}$ and the realized removed subspace $\widehat{\mathcal{U}}$. This model admits robust common modes, fixed-template regression, and masked or weighted low-rank families while requiring their distinct fit and application rules.

1.2 Correlation is not a physical label

Detector correlation establishes shared structure, not physical origin. Atmosphere can be important, but electronics, calibration residuals, detector noise, astronomical structure, or mixtures can occupy the same statistical subspace. An array-wide component is a detector-membership and geometry statement, not proof of an atmospheric origin. A component visible in conditioned r is not automatically electronics-only.

Subtraction follows the admitted statistical object, not its informal name. Sky that overlaps the removed subspace is attenuated even when the mode was called atmosphere. The contract therefore publishes the fitted component, subspace, supports, and response while leaving physical attribution diagnostic unless separately established.

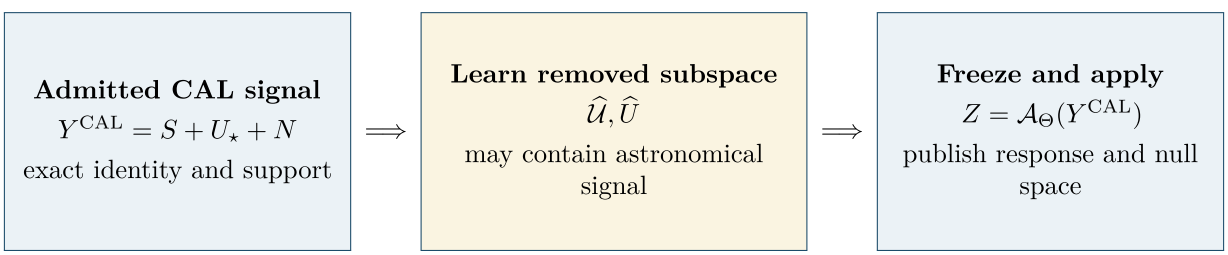

Figure 1: Scientific flow of correlated-mode cleaning. The removed subspace is part of the result because statistical correlation does not identify physical origin. This schematic is explanatory; the formal operator contract is in the engineering specification.

1.3 The removed subspace is part of the result

The transformed timestream alone is incomplete. PTC must identify the removed subspace, metric, rank, support, grouping, fit order, centering/scaling, degeneracy, gauge, and exact application map. Basis sign, reciprocal basis/loading scale, component order, and rotations within a degenerate space are representation freedoms. Scientific claims attach to the modeled correlated signal and removed subspace.

The null space includes perturbations made unidentifiable by the complete operation. A detector-common sky mode can be removed while output retains the same unit. Centering can remove an additive reference. A source mask can protect its registered model while leaving unmasked extended emission exposed. Retaining mJy/beam is therefore not a response claim.

2 Validity is use-specific support

2.1 A cause does not prescribe one action

Each typed cause maps independently to basis fitting, loading fitting, application, output retention, coefficient/QC estimation, response, empirical estimation, simulation, and downstream use. One occurrence can be excluded from basis learning, admitted for frozen application, retained in output, and excluded from a coefficient. An output-only rejection does not retroactively change the learned subspace.

Only eligible finite occurrences enter arithmetic. Missing and non-finite remain distinct from numerical zero. Replacing an excluded value by zero before ordinary PCA changes means, covariance, singular values, and possibly rank. A weighted missing-data estimator can assign zero statistical influence, but that is a declared property of that estimator [1].

One occurrence with cause ϕ	Basis fit	Frozen application	Output	Coefficient
Example policy	Excluded	Admitted	Retained	Excluded
Scientific meaning	No learning influence	Receives declared projection	Value may remain	No coefficient influence

Figure 2: Use-specific support. A flag records a cause; the immutable policy maps that cause separately into each scientific use. No single good/bad bit can encode this example.

2.2 Direct validity and causal influence

Direct validity describes an occurrence itself. Causal influence records what could have affected an output through a fit, selection, classification, or coefficient. One sample used to learn a common mode can influence many other detector-time outputs even where its local numerical derivative is zero. Direct synthesized or RTC-replaced occurrences remain excluded from the calibrated science branch; noncenter influence is preserved and routed by use-specific policy.

3 Coordinates, grouping, and source protection

3.1 Centering, scaling, and gauge define the operator

Centering each detector over time is not centering detectors at each time. Scaling by detector scatter is not scaling by a formal noise model. Every coordinate transform therefore declares its axis, population, support, weights, estimator, boundaries, unit, inverse, gauge, and null-space consequence. Internal invertible scaling is restored before ordinary output. Noninvertible transformation changes the signal definition and needs a separately named product role.

3.2 Within-array hierarchy

Base v0.1 fits within one TolTEC array. A hierarchy may contain an array-wide component, network/electronics components, and optional local or focal-plane components. Joint and sequential fitting are generally different when subspaces overlap; the exact order and cumulative removed subspace are state. Data-derived groups retain their population, metric, candidate rule, cardinality controls, stability evidence, and selection uncertainty.

Cross-array fitting remains unavailable without separate spectral, calibration, beam, source-color, and response authority. Equal unit labels cannot supply those missing physical relations.

3.3 External source/residual parents have finite authority

A source model, mask, residual, or prior-pass parent may be consumed only with exact owner, unit, frame, registration, validity, response, support, boundary, and recurrence identity. A mask protects only its declared model and support. It does not prove survival of unmasked compact emission, extended emission, or other sky components correlated with the fitted subspace. Missing recurrence or registration identity makes the affected protected/fitted role unavailable, not approximately safe.

4 Learn, select, resolve, and apply

4.1 Estimator families and the one-way lifecycle

The lifecycle is:

$$\begin{aligned}&\text{request} \to \text{effective policy} \to \text{observation-resolved inputs} \to \text{learned evidence}\\ &\qquad \to \text{resolved model} \to \text{application} \to \text{publication.}\end{aligned}$$

Learned evidence contains every evaluated candidate, its fitted state, diagnostics, predicate values and causes, ordering, ties, and disposition. The selected resolved model is immutable and contains the exact application map. Application and publication are later states; a published rank is not a substitute for the evidence explaining why it was selected.

Admitted base families remain distinct. A robust common-mode family declares its location estimator, aggregation weights, denominator, population, loadings, and robust-loss behavior. A fixed-template regression declares template parent, metric, generalized inverse, tolerance, and supports. A masked/weighted PCA/SVD family declares its low-rank objective, missing-data semantics, metric, candidate construction, rank, degeneracy, and exact application rule.

These choices also carry different scientific strengths. A robust group common mode is comparatively interpretable and resistant to isolated outliers, but can be inadequate when several correlated structures coexist. Fixed-template regression is appropriate when a physically or diagnostically motivated temporal basis already exists, but its authority is limited by the supplied model and parent. PCA/SVD can capture several unknown mixtures, but its learned subspace can contain sky and depends strongly on metric, grouping, support, and rank. Conditioned-r PCA supplies electronics-enriched diagnostic evidence; it neither establishes electronic origin nor controls the calibrated-x branch in base v0.1.

4.2 The exact frozen application map

The owner-approved base-v0.1 PCA/SVD rule is projection of any admitted application input through the frozen realized subspace and metric. It is not subtraction of one frozen numerical reconstruction. The realized family must say which space contains the removed basis and whether application is a temporal-left action, detector-right action, two-sided action, detector/time- specific action, or an exact vectorized operator. It must also state which quantities are frozen and which coefficients are recomputed linearly during application.

This resolves a response-critical ambiguity. On the fitted parent, frozen-component subtraction and frozen-subspace projection can both equal $Y^{\mathrm{CAL}} - \widehat{U}$. On a perturbation H, however, frozen-component subtraction has identity derivative, whereas frozen-subspace projection removes the component of H in the selected space. Likewise, temporal-left and detector-right projection generally act differently on H.

Frozen-component subtraction remains a possible separately named affine family, with identity derivative, but it is not silently called PCA/SVD projection and is not the base-v0.1 PCA application rule.

A fit-excluded occurrence is application-available only when the resolved frozen operator defines its value from admitted fitted state and support. If a required basis coefficient or coefficient-recomputation input, neighboring support, detector binding, metric term, coordinate transform, or boundary state is unavailable, application at that occurrence is unavailable rather than implicitly interpolated, zero-filled, reconstructed, or extended from the fit population.

4.3 Least-aggressive conjunctive selection

Candidates are finite and ordered from least to more aggressive. The first candidate is admitted only when every required residual-contamination, astronomical-transfer, conditioning, support, stability, and QC predicate passes. A failed transfer predicate cannot be compensated by residual suppression or a scalar score. Nonnested candidates are compared through their complete subspaces and responses, not mode count alone.

The astronomical-transfer predicate is itself typed: source class and companion, fixed-state or full-procedure response, amplitude/location domain, metric, tolerance, mask/support, upstream response, uncertainty treatment, and exact tested claim. A transfer number without those identities is not a candidate-admission result.

4.4 Diagnostic-only conditioned r

A separately authorized conditioned r product can support diagnostic PCA. In base v0.1 it remains inert or advisory. Changing r cannot change calibrated-x fit membership, subtraction, output, or coefficients, and it cannot supply an x subtraction basis.

5 Post-fit assessment and immutable-parent refinement

5.1 What diagnostics can and cannot establish

Relevant diagnostic families include residual PSD or variance, coherence, removed variance, loading magnitude and stability, leverage, protected-source response, non-Gaussianity, transients, and conditioned-r comparison. Each needs an estimator, population or approved model, normalization, support, uncertainty, and policy role.

A population outlier is not automatically detector pathology. Astronomical signal, source-model failure, calibration state, focal-plane position, and expected sensitivity variation are competing explanations. Numerical classification thresholds remain owner-controlled inputs.

5.2 Refit from the immutable CAL parent

When classification changes fit support, the complete selected model is refitted from the same immutable admitted CAL parent. A cleaned output is not the next fit parent. The final resolved model is applied once. This prevents detector-QC refinement from becoming accidental repeated cleaning with compounded subtraction and null spaces.

Sequential residual fitting is permitted only as one explicit ordered hierarchical estimator with cumulative subspace, response, covariance, and parentage. The refinement loop is finite and distinguishes stable state, no-new-threshold event, maximum refinements, oscillation, insufficient support, nonconvergence, exhaustion, and error.

5.3 Internal iteration, PTC pass, and FRUIT recurrence

Internal numerical iteration solves one fit. A new PTC pass has a new immutable input parent and realized identity. External FRUIT recurrence owns source-model subtraction/add-back, map feedback, restart, and recurrence parentage. These identities cannot be interchanged.

6 Coefficient types, uncertainty, and bias

6.1 A loading is not a weight

A fitted loading couples a detector to a mode/template and changes under basis gauge. A centering/scaling parameter defines coordinates. A diagnostic summarizes a declared statistic. Only an explicitly named analysis/gridding family may be MAP-facing. That family declares whether its index is detector or detector-time, its statistic and factor domains, unit, normalization operator and domain, support, lifecycle, consumers, and prohibited meanings.

An empirical-scatter coefficient is not automatically precision. Re-estimating a coefficient creates new realized state and does not silently modify an earlier transformed signal.

6.2 Conditional, selection, systematic, and bias claims

Conditional covariance propagates through the derivative or affine linear part of the exact frozen application map. It can contain temporal and inter-detector off-diagonal terms. When rank, groups, masks, classifications, or coefficients vary, total uncertainty includes between-selection variation. Omitting that term is a conditioned approximation, not complete uncertainty.

Formal covariance, marginal variance, empirical scatter, calibration/systematic uncertainty, response uncertainty, selection uncertainty, and cross-coordinate covariance remain separate types. An absent cross-observation block means unknown, not independent.

Each family and product role also states its conditional expectation or bias claim for astronomical signal, fitted nuisance, remaining detector noise, calibration uncertainty, and selection. If a claim is not supported, it is unavailable. A single zero-mean synthetic residual cannot establish general unbiasedness.

7 Response domains: local, chain, and full procedure

7.1 Local operators and propagated companions

Three objects are distinct:

1. the PTC-local derivative or linear part of the frozen application map;
2. the complete chain response formed by composing that local operator with the complete admitted upstream response ending on the CAL grid; and
3. the realized propagated detector-time companion.

A source-domain template is first carried through the complete admitted upstream response ending on the CAL grid, then through the PTC-local operator. That upstream response is the response carried by the CAL product, including every applicable admitted source-to-detector relation, beam convention, coordinate and scan geometry, RTC operation, and CAL operation; it is not only the CAL-owned multiplicative operator. If an exact detector-time companion is already supplied on the CAL grid, propagation begins directly with the PTC-local operator. The upstream response is never applied twice.

Typed response chain. $\tau_{\mathrm{src}} \xrightarrow{K_{\mathrm{up}\to\mathrm{CAL}}} k_{\mathrm{src}}^{\mathrm{CAL}} \xrightarrow{J_\Theta[Y]} k_{\mathrm{src}}^{\mathrm{PTC}|\Theta}$. An already realized $k_{\mathrm{src}}^{\mathrm{CAL}}$ enters at the middle node.

7.2 Full-procedure signal response and state change

Fixed-state response asks what the resolved operator does while learned state is held fixed. Full-procedure response reruns learning, selection, rank, classification, refinement, and application from the immutable CAL parent. Its finite difference is defined only on the transformed-signal projection of the procedure. Masks, ranks, groups, classes, and stop causes are not numerically subtracted; their changes are retained through a typed state comparison.

Perturbation amplitude, side, source state, location, and domain are part of the response. If response changes materially across them, the product is a bounded family, not one global kernel. A whole-chain RTC-to-CAL-to-PTC injection is separate again because PTC alone cannot relearn upstream state [3].

7.3 The optional map-center diagnostic

The optional map-center point-source diagnostic names whether it uses a fixed-state or full-procedure companion. It also requires the exact source parent, propagated companion, reference MAP functional, center selector, band/mode/state binding, and uncertainty status. It does not confer general PTC, off-center, extended-source, arbitrary-morphology, beam-shape, or MAP authority.

8 Products, surrogates, state axes, and boundaries

8.1 The authoritative intermediate and complete provenance

The in-memory transformed TOD is the authoritative PTC-to-MAP intermediate on the PTC-dependent route. A persisted copy declares diagnostic-artifact or requested-analysis role. Persistence grants no map, beam, significance, archival, validation, or production authority.

Material provenance follows the full one-way lifecycle: immutable parent, request, effective policy, observation-resolved inputs, learned evidence, resolved model, application, publication, supports, subspace, coefficients, classification, response, covariance, randomness, and output identity. Required siblings and links are atomic; a readable partial file is not a complete product.

8.2 Shifted/null surrogates and insufficiency

A shifted or null surrogate is a controlled diagnostic or simulation population used to estimate false associations, a null distribution, or empirical variability; it is not additional astronomical exposure. A shifted or null surrogate moves signal and its associated validity/eligibility, support, boundaries, randomness, and parent identity together. Shifting values while leaving membership fixed changes the scientific population and is not the declared surrogate. Insufficient transformed support is unavailable or rejected, never a valid numerical-zero fallback.

Every fallback records cause, stage, changed operation, and response/covariance consequences. Randomness that can affect science records generator/version, seed, stream, draw identity, and input identity.

8.3 Independent state axes and ownership boundaries

Product realization, response availability, covariance availability, and validation/evidence disposition are independent axes. A realized transformed signal may have unavailable complete response. A disabled PTC route has no realized PTC signal even when an upstream CAL response exists. Not produced is therefore not a response-availability value.

PTC does not own RTC conditioning, CAL calibration/atmosphere, AST/ALIGN coordinates, VAL reusable eligibility, MAP estimation, NOI uncertainty calibration, BEAM interpretation, FLT filtering, or FRUIT recurrence. Base v0.1 also excludes raw-$\Delta f/f$ Beammap PTC, raw-r numerical science, r-derived x subtraction, unconstrained joint x/r PCA, cross-array fitting, and polarimetry.

9 Scientific validation program, evidence status, and owner decisions

9.1 Scientific validation program

Before numerical thresholds or production dispositions are selected, the science program should declare the following study families and observables. This table specifies studies, not achieved results or acceptance thresholds.

Study	Variations	Measurements
Missing data and flags	Missingness pattern, zero fill, weighted exclusion, fit-excluded/apply-allowed	Fitted subspace, rank, transformed signal, bias, and use-specific support
Estimator family	Robust common mode, fixed template, and PCA/SVD	Residual coherence, null modes, source transfer, and covariance
Group hierarchy	Array, network, and local groups; joint versus sequential order	Cumulative removed subspace, order dependence, residuals, and sky response
Mode admission	Candidate ranks/subspaces, source classes, support, and noise regimes	Selected candidate, stability, residual contamination, and compact/extended response
Detector refinement	Leverage, residual outliers, threshold changes, and oscillating membership	Final detector set, refit stability, bias, and data loss
Response	Fixed-state versus full-procedure; amplitude, side, location, and mask boundary	Propagated companion, typed state changes, nonlinearity, and response family
Conditioned r	Optical leakage, electronics-common modes, and tune/loading state	Diagnostic-mode stability, x/r correspondence, and false physical attribution
Surrogates	Shifted/null membership, support, boundary, and random stream	Null distribution, false detections, and membership fidelity

9.2 Evidence layers

Layer	Status of this draft
Algebraic contract	Defined and mechanically traceable across the shared formal core.
Implementation conformity	Unassessed; no implementation evidence entered this revision.
Representation and response fidelity	Not performed; later evidence must bind serialized state to the realized numerical operator.
Observational performance	Not performed; no transfer, residual, covariance, or source-recovery threshold is claimed achieved.
Production readiness	Not assessed and not authorized by this document.

9.3 Decision and availability register

Item	State	Consequence
Conditioned r use	Decided	Diagnostic-only and inert/advisory for base v0.1; cross-channel subtraction is unavailable.
PCA/SVD frozen application	Decided	Project through the frozen realized subspace and metric; exact acting space is family state.
Numerical selection predicates	Open policy input	Automatic candidate admission is unavailable until every required typed predicate is supplied.
Mandatory/default estimator	Open policy input	An absent estimator request cannot be replaced by an inferred default.
Centering/scaling choices	Open policy input	Concrete coordinates must be supplied for each product role.
Detector diagnostic thresholds	Open policy input	Policy-changing classification requires registered thresholds.
Conditioned r producer	Known, not supplied	The optional diagnostic branch remains unavailable.
Reference MAP functional	Known, not supplied	The optional map-center diagnostic remains unavailable.

Item	State	Consequence
Complete covariance model	Deferred	Precision/significance claims remain blocked.
Full-procedure perturbation domain	Open policy input	A concrete response family needs amplitude, side, source state, location, and validity domain.
Implementation and validation evidence	Not assessed	No conformity, achieved performance, freeze, or readiness claim follows.

A Compact rationale-to-contract crosswalk

The exact identifier-by-identifier crosswalk is supplied with the package. This compact view shows where the principal scientific questions are governed in the companion engineering specification.

Scientific question	Rationale location	Formal authority
What is estimated and removed?	Sections 1 and 3	Definitions 001–008; Requirements 019–025
How is validity routed?	Section 2	Definitions 009–013; Requirements 011–018
What is learned, selected, resolved, and applied?	Section 4	Definitions 014–015, 029–032, and 040; Requirements 026–040, 083–085, and 089
How does detector refinement repeat?	Section 5	Definitions 019–021; Requirements 042–051
What do coefficients and uncertainties mean?	Section 6	Definitions 023–024 and 039; Requirements 052–060 and 086
How are response domains separated?	Section 7	Definitions 016–018, 033–034, and 041; Requirements 061–068 and 087
What are product and surrogate states?	Section 8	Definitions 025, 028, 035–037; Requirements 069–079, 081–082, and 088
How will realizations be studied?	Section 9.1	Predictions 001–049
What evidence is claimed?	Section 9.2	Requirement 080 and Prediction 038

B Provenance and program adherence

SCI-PTC follows the Citlali Scientific Contract Library Program and began from an owner-reviewed prior-work recovery record rather than a fresh derivation. The implementation-blind author packet reused an independent mathematical core under a binding supersession cover and excluded implementation, audit, repair, test, validation, Unity, and production material. The r0.3 revision implements the bounded scientific-owner review of r0.2 after the explicitly approved frozen-subspace projection decision. It does not reopen mature numerical algorithms or make an implementation claim.

References

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